



Data Article

CPSMI2025: A curated dataset of conventional Pap smear microscopy images for deep learning-based cervical cancer screening



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ABSTRACT

Cervical cancer remains one of the most common and deadly cancers in low-resource regions, where conventional Pap smear screening - though affordable - faces diagnostic bottlenecks due to limited specialist availability. To support the development of automated screening tools, we present CPSMI2025: a curated dataset of 2169 high-resolution Pap smear microscopy images representing nine clinically relevant cytology categories, derived from >350 manually screened slides obtained from Hospital General de Zona No 2 (IMSS) and a private pathology practice in Tuxtla Gutiérrez, Chiapas, Mexico. All slides were pre-classified by both a pathologist and cytotechnologist. Images were cap-

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tured using a replicable open-source low-cost microscopy platform.

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Specifications Table

Subject	Health Sciences, Medical Sciences & Pharmacology
Specific subject area	Cervical cancer, Cervical cancer screening, Conventional Papanicolaou smear, Image classification, Computer vision
Type of data	Images
Data collection	Over 350 conventional cytology slides, each from a distinct patient, were manually scanned to identify fields of view (FoVs) containing cells representative of nine defined categories relevant to cytologic analysis. These slides had been previously classified by both a cytotechnologist and a pathologist. Images were captured using a custom, replicable microscope (described in [1]) with a 20x Newport objective. While each image could contain multiple cells of varying types, abnormal cells were classified based on the most advanced lesion present, whereas images with normal cells were categorized according to the most abundant cell type. The slides were provided by Hospital General de Zona No 2 of the Instituto Mexicano del Seguro Social (IMSS) and a private pathology practice, both in Tuxtla Gutiérrez, Chiapas, México.
Data source location	Hospital General de Zona No 2, Instituto Mexicano del Seguro Social (IMSS) in Tuxtla Gutiérrez, Chiapas, México.
Data accessibility	Dr. Tobar's private practice, Tuxtla Gutiérrez, Chiapas, México. Repository name: CPSMI2025 Direct URL to data: https://data.mendeley.com/datasets/rmfyhrtfkfw/1 DOI: 10.17632/rmfyhrtfkfw.1
Related research article	J. Ocampo-López-Escalera, H. Ochoa-Díaz-López, X.M. Sánchez-Chino, C.A. Irecta-Nájera, S.D. Tobar-Alas, M. Rosete-Aguilar, A low-cost platform for automated cervical cytology: addressing health and socioeconomic challenges in low-resource settings, <i>Front. Med. Technol.</i> 7:1531,817 (2025). https://doi.org/10.3389/fmedt.2025.1531817

1. Value of the Data

- This dataset consists of 2169 conventional Pap smear microscopy images spanning nine cytologic categories grouped into four major categories relevant for cervical cancer screening. It covers a range of cellular changes, from normal morphologies to malignant lesions, providing a robust resource for developing automated screening tools.
- All images were captured using a low-cost and replicable microscope. The public nature of this system allows other research groups to expand the dataset under identical parameters, facilitating the inclusion of additional cellular types or increasing image counts while maintaining consistency. This approach particularly supports the incorporation of rare but clinically significant cell types that may be underrepresented in current collections.
- The dataset holds particular relevance for low-resource settings where cervical cytology remains the primary screening method due to its cost-effectiveness and minimal infrastructure requirements compared to more sophisticated screening techniques. By focusing on conventional Pap smear images, this collection directly addresses the need for AI tools compatible with the screening methods most widely used in resource constrained health programs.
- Unlike most publicly available datasets that focus on isolated cell images, this collection preserves full fields of view (FoVs) as they appear during actual microscopic examination. Each image may contain anywhere from a single cell to dozens of cells, including complex overlapping structures that mirror real-world screening conditions.
- Researchers may use this dataset to train and validate deep learning models for automated cervical cytological analysis. The full FoV images provide a robust foundation for developing algorithms capable of handling certain degree of variability encountered in clinical practice.

2. Background

Cervical cancer is still one of the most common and deadly cancers worldwide, despite being highly preventable and highly curable when detected in early stages. In 2022, cervical cancer took the lives of nearly 350,000 women worldwide, with >90 % of deaths occurring in low- and middle-income countries [2]. One of the main reasons for this disparity is the lack of effective infrastructure for early detection in low-resource settings.

Unlike most cancers, cervical cancer has a well-established cause: a persistent infection with high-risk strains of the human papillomavirus (HPV). Without effective treatment, these infections could progress to cancer over time. This progression typically occurs over 10 to 20 years, during which precancerous cellular changes (dysplasia)—detectable under a conventional microscope—develop [3]. Regular screening helps to identify these abnormalities before they become cancerous and to treat them properly. In high-income countries, modern screening methods—such as the human papillomavirus (HPV) test—have been implemented, demonstrating high sensitivity and specificity. However, deploying these tests in low-resource settings is often unfeasible due to their high cost and the need for advanced laboratory equipment and trained personnel [4]. On the other hand, conventional Pap smear is far more affordable and requires minimal infrastructure, making it the primary screening method for cervical cancer in resource-limited regions. However, it still relies on specialists to manually analyze samples under a microscope, creating a diagnostic bottleneck.

Deep learning (DL) algorithms have emerged as a promising solution for automating slide analysis, reducing dependence on specialists [5]. Yet, training these models requires large, pre-classified datasets. The dataset presented in this paper complements existing public datasets while offering distinct advantages.

3. Data Description

This dataset comprises 2169 Pap smear microscopy images organized into nine subclasses further grouped within four major classes (Fig. 1). The subclasses represent some of the most clinically relevant cellular types in cytological analysis, as detailed in Table 1, where each subclass is briefly described and exemplified with a representative image.

While not all subclasses directly correspond to categories in The Bethesda System (TBS) [6]—the most widely adopted framework for reporting cytological findings—each can be meaningfully

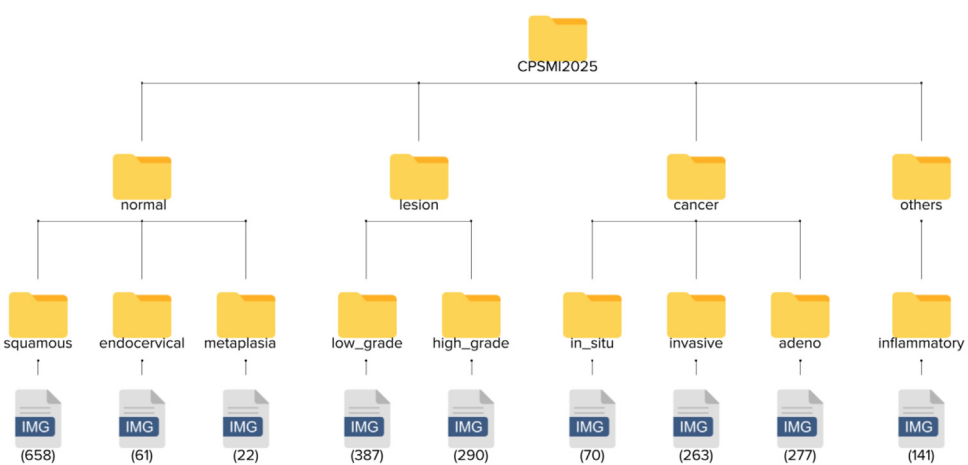
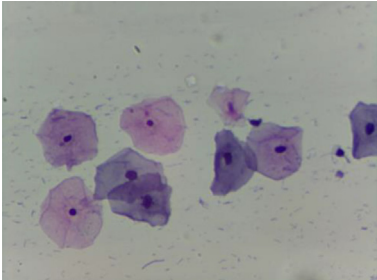
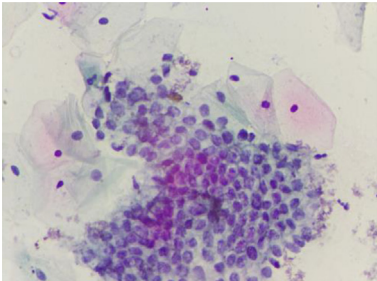
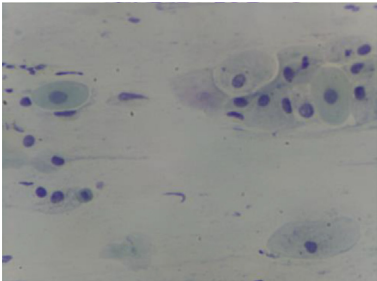
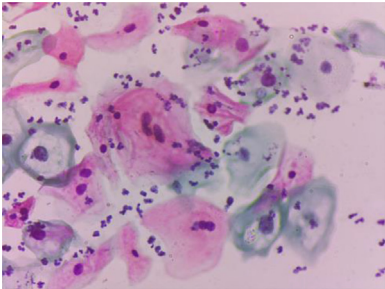


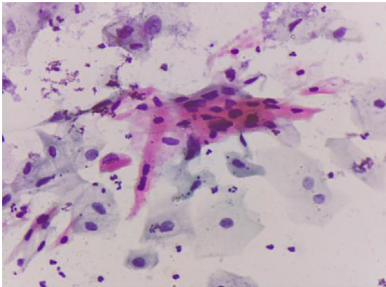
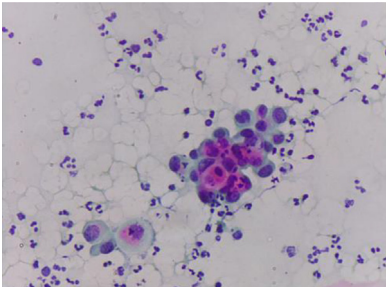
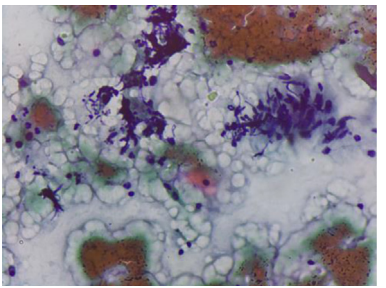
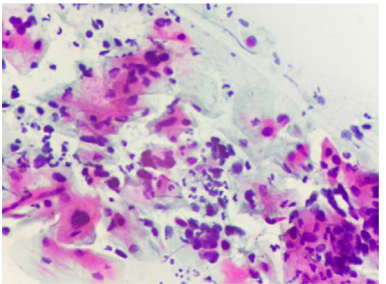
Fig. 1. Dataset directory structure and number of images across diagnostic classes.

Table 1
General characteristics and sample images of the 9 subclasses that compose this dataset [6–8].

Major class/Subclass/No. of images	Description	Sample image (FoV real size: 314.34 μm x 235.6 μm)
Normal/ Squamous/ 658	The squamous subclass includes the two predominant cell types in a normal cytologic smear in women of reproductive age: superficial and intermediate. Superficial cells represent the most mature layer of the stratified squamous epithelium. They contain a small, pyknotic nucleus (5–6 μm) and abundant polygonal cytoplasm (45–50 μm), which typically stains pale pink or green with Pap stain. The nuclear-to-cytoplasmic (N/C) ratio is low (2–3 %). Intermediate cells, in comparison, display a larger nucleus (≈8 μm) with finely granular chromatin, surrounded by polygonal cytoplasm measuring 35–40 μm. The cytoplasm stains pale pink or green, and the N/C ratio is 3–5 %.	
Normal/ Endocervical/ 61	Endocervical cells are mucin-secreting columnar cells of glandular origin. They usually appear in strips (picket-fence pattern) or flat sheets (honeycomb arrangement) rather than as isolated cells. Their cytoplasm is abundant, vacuolated, and lightly stained, measuring about 15 μm. The nucleus is ≈8 μm with finely granular chromatin.	
Normal/ Metaplasia/ 22	Metaplastic cells arise from the transformation of glandular endocervical cells into squamous epithelium, a normal and non-neoplastic process occurring at the transformation zone, often under hormonal influence (e.g., estrogen) or local irritation. They may appear with immature or mature features, but in both forms the nuclei remain relatively uniform in size and shape, with a higher nuclear-to-cytoplasmic (N/C) ratio than in fully differentiated squamous cells. Immature metaplastic cells can sometimes mimic atypical squamous cells; however, the absence of nuclear pleomorphism, or irregular chromatin helps to distinguish them from dysplastic changes.	
Lesion/ Low grade/ 387	Low-grade squamous intraepithelial lesion (LSIL) is characterized by large squamous cells resembling intermediate or superficial types. They exhibit nuclear enlargement (≈3 × intermediate nuclei), hyperchromasia, mild irregularities of nuclear membranes, and/or uniform chromatin. Binucleation or multinucleation is common, while nucleoli are usually absent. A hallmark feature is koilocytosis, a perinuclear cytoplasmic clearing with a dense rim, caused by HPV infection. Despite atypia, LSIL cells preserve polygonal morphology.	

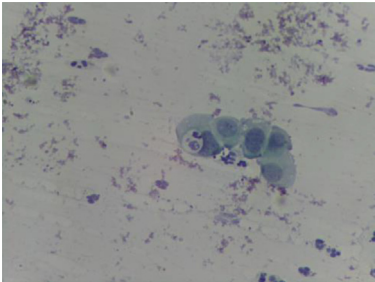
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Table 1 (continued)

Major class/Subclass/No. of images	Description	Sample image (FoV real size: 314.34 μm x 235.6 μm)
Lesion/ High grade/ 290	High-grade squamous intraepithelial lesion (HSIL) is represented by smaller squamous cells, in comparison to LSIL cells, with markedly increased N/C ratios. Cells may occur singly, in sheets, or as syncytial clusters. Nuclei are hyperchromatic with irregular contours (grooves or indentations), and nucleoli are typically absent.	
Cancer/ In-situ/ 70	Carcinoma in situ is confined to the squamous epithelium and appears in two variants: nonkeratinizing and keratinizing. Nonkeratinizing: Crowded groups or syncytial fragments of medium-to-large cells with high N/C ratios, irregular nuclear contours, coarse chromatin, prominent nucleoli, and occasional naked nuclei. Keratinizing: More dispersed cells with hyperchromatic, irregular granular chromatin, rare nucleoli, and keratinized cells showing orangeophilic cytoplasm. Squamous pearls and tadpole-shaped cells may also be present.	
Cancer/ Adenocarcinoma/ 277	Adenocarcinoma involves malignant transformation of glandular cells and may be endocervical or endometrial in origin. Endocervical type: Pleomorphic cells, with fluffy cytoplasm, vacuolization, true nuclear crowding, loss of polarity, and nuclei showing coarse chromatin and clumped aggregates. Endometrial type: Single cells or small clusters with nuclear size variation (depending on grade), polarity loss, hyperchromasia, chromatin clearing in higher grades, increasing nucleolar prominence, scant vacuolated cytoplasm, intracytoplasmic neutrophils, and watery granular tumor diathesis.	
Cancer/ Invasive/ 263	Invasive carcinoma is characterized by malignant cells breaching the epithelial basement membrane and infiltrating the stroma. Both squamous and glandular components may be affected. The defining cytologic feature is marked atypical nucleomegaly in association with evidence of stromal invasion.	

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Table 1 (continued)

Major class/Subclass/No. of images	Description	Sample image (FoV real size: 314.34 μm x 235.6 μm)
Other/ Inflammatory/141	This subcategory includes fields of view associated with infectious organisms. While such infections and their effects may mimic dysplastic changes, they lack true nuclear atypia and are often accompanied by inflammatory infiltrates. Common organisms include: Trichomonas vaginalis, Chlamydia trachomatis, Herpes simplex virus (HSV), Candida species, Bacteria: Gardnerella vaginalis, bacilli, leptothrux, Actinomyces. Additionally, abnormal lymphocytic populations related to infection may also be observed.	

associated with a TBS class. These subclasses were defined based on morphological characteristics and image availability, ensuring they contain distinguishable features suitable for classification by artificial intelligence models.

All images share consistent technical specifications: 3 channel JPG format with dimensions of 2028 × 1520pixels.

4. Experimental Design, Materials and Methods

The image dataset was curated using >350 cytology slides, each from a distinct patient, previously classified by both a pathologist and a cytotechnologist. These slides were sourced from *Hospital General de Zona No 2* of the *Instituto Mexicano del Seguro Social (IMSS)* and Dr. Tobar's private practice, both located in Tuxtla Gutiérrez, Chiapas, México. Including two sources allows to achieve certain variation in staining techniques, which is a positive characteristic for generalization purposes.

Images were captured using a replicable, automated low-cost microscope (described in [1]), equipped with a Newport MVC-20X objective offering 20 × magnification, 0.4 numerical aperture, and 8.72 mm working distance. The microscope also integrates an HQ Raspberry Pi camera featuring a 12.3-megapixel Sony IMX477 sensor. To account for variable lighting conditions caused by differences in cell density across fields of view, images were captured at full sensor resolution with auto-exposure adjustment. Subsequently, they were downsampled to one-quarter of their original dimensions using Python's Pillow library, employing the LANCZOS filter for high-quality resampling.

During curation, slides were manually scanned to identify fields of view FoVs representative of the previously described nine categories. Each FoV could contain multiple cell types; abnormal images were classified based on the most advanced lesion present, while normal images were categorized by the most abundant type. A custom Python script facilitated image capture, saving each FoV to a subclass-specific folder based on keyboard input.

A post-curation quality check was performed to eliminate severely out-of-focus or duplicated images. It is important to highlight that slightly out-of-focus images were intentionally retained to simulate real-world conditions where perfect focus may not always be achievable by autofocus algorithms, ensuring the dataset's practical relevance.

Limitations

- The diversity of cellular and non-cellular elements in cervical cytology is vast. However, the current dataset does not encompass the full spectrum of images typically encountered during a full Pap smear slide review.
- The nine subcategories encompassed in the repository are defined on the basis of cytologic characteristics as well as the final slide diagnosis. It is important to note that cytology is a complex discipline that, in practice, integrates information from multiple fields of view to reach a diagnosis. As a result, the classification of an individual FoV can be challenging and, at times, may appear somewhat arbitrary between particular classes. However, for this reason, the nine subcategories are consolidated into broader, more robust major classes.
- Additionally, the dataset exhibits *class imbalance* across diagnostic subcategories. For instance, the *metaplasia* class is underrepresented (22 images) compared to the *squamous* class (658 images). Such imbalances should be accounted for during model training to avoid skewed performance.

Ethics Statement

This study was reviewed and approved by the Research Ethics Committee of El Colegio de la Frontera Sur (approval number CEI/1022/2025) in compliance with the Declaration of Helsinki. The human samples were obtained from a diagnostic pathology laboratory without disrupting its regular workflow. Written informed consent was not required given that:

1. Full confidentiality was ensured, researchers accessed only de-identified samples and their associated diagnostic data.
2. No personal identifiers were available to the study team at any stage.
3. Sample use did not affect clinical care or laboratory workflows.

CRedit Author Statement

José Ocampo-López-Escalera: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing - Original Draft, Visualization. **Martha Rosete-Aguilar:** Conceptualization, Resources, Writing - Review & Editing, Supervision, Project administration. **Héctor Ochoa-Díaz-López:** Conceptualization, Resources, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Xariss M. Sánchez-Chino:** Resources, Writing - Review & Editing, Supervision. **César A. Irecta-Nájera:** Resources, Writing - Review & Editing, Supervision. **F. Guillermo Domínguez-Gómez:** Data curation, Writing - Review & Editing. **Saúl D. Tobar-Alas:** Data curation, Writing - Review & Editing.

Data Availability

[CPSMI2025 \(Original data\)](#) (Mendeley Data).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] J. Ocampo-López-Escalera, H. Ochoa-Díaz-López, X.M. Sánchez-Chino, C.A. Irecta-Nájera, S.D. Tobar-Alas, M. Rosete-Aguilar, A low-cost platform for automated cervical cytology: addressing health and socioeconomic challenges in low-resource settings, *Front. Med. Technol.* 7 (2025) 1531817, doi:[10.3389/fmedt.2025.1531817](https://doi.org/10.3389/fmedt.2025.1531817).
- [2] World Health Organization, International Agency for Research on Cancer (IARC), Cancer Today (GLOBOCAN 2024) (2024) [Online]. Available: <https://gco.iarc.fr/today/en> . [Accessed: July, 1, 2025.].
- [3] World Health Organization Comprehensive Cervical Cancer Control: A Guide to Essential Practice, 2nd ed., WHO Press, Geneva, 2014 Available at: <https://www.ncbi.nlm.nih.gov/books/NBK269623/> .
- [4] R. Gupta, S. Gupta, R. Mehrotra, P. Sodhani, Cervical cancer screening in resource-constrained countries: current status and future directions, *Asian Pac. J. Cancer Prev* 18 (6) (2017) 1461–1467 PMID: 28669152; PMCID: PMC6373785, doi:[10.22034/APJCP.2017.18.6.1461](https://doi.org/10.22034/APJCP.2017.18.6.1461).
- [5] T. Conceição, C. Braga, L. Rosado, M.J.M. Vasconcelos, A review of computational methods for cervical cells segmentation and abnormality classification, *Int. J. Mol. Sci.* 20 (2019) 5114, doi:[10.3390/ijms20205114](https://doi.org/10.3390/ijms20205114).
- [6] R. Nayar, D.C. Wilbur, *The Bethesda System for Reporting Cervical Cytology: Definitions, Criteria, and Explanatory Notes*, Springer, Cham, 2015, doi:[10.1007/978-3-319-11074-5](https://doi.org/10.1007/978-3-319-11074-5).
- [7] M. Kamal, F. Topiwala, Nonneoplastic cervical cytology, *Cytojournal* 19 (2022) 25 PMID: 35510107; PMCID: PMC9063585, doi:[10.25259/CMAS_03_06_2021](https://doi.org/10.25259/CMAS_03_06_2021).
- [8] E.S. Cibas, Chapter 1 - cervical and vaginal Cytology, in: E.S. Cibas, B.S. Ducatman (Eds.), *Cytology*, Fourth Edition, W.B. Saunders, Philadelphia, 2014, pp. 1–57. ISBN 9781455744626. <https://doi.org/10.1016/B978-1-4557-4462-6.00001-5>.